

SMART ATTENDANCE SYSTEM

A Complete Implementation Guide: Face Recognition & Hardware Integration

1. Introduction

This report provides a step-by-step blueprint for building a **Smart Attendance System**. The system uses computer vision to identify students and automatically logs their presence in an Excel sheet while interacting with physical hardware (Arduino) for real-time feedback.

2. Requirements

Software Components

- **Python 3.x:** The primary programming language.
- **OpenCV:** For camera handling and image processing.
- **DeepFace:** For high-accuracy facial recognition.
- **Pandas:** To manage and save attendance data in Excel format.
- **PySerial:** To communicate between Python and Arduino.

Hardware Components

- **Webcam:** To capture real-time video.
- **Arduino Uno:** The microcontroller to handle physical signals.
- **LEDs/Buzzer:** For visual/audio confirmation of attendance.
- **Jumper Wires & Breadboard:** For circuit connections.

3. System Architecture

The system works in a loop: **Capture Image** → **Process with DeepFace** → **Match with Database** → **Send Signal to Arduino** → **Update Excel Log**.

4. Step-by-Step Implementation

Step 1: Database Setup

Create a folder named dataset. Inside this folder, save one clear image of each student. Rename the image files to the student's name (e.g., Mustafa.jpg). The system will use these filenames as the identity.

Step 2: Hardware Wiring

Component	Arduino Pin	Description
Green LED	Pin 13	Glows when attendance is marked.
Buzzer	Pin 12	Beeps on successful recognition.
Ground (GND)	GND	Common ground for all components.

Step 3: The Python Logic

The Python script initializes the camera and loads the DeepFace model. For every frame, it calculates the distance between the captured face and the database. The logic follows:

```
# Logic Snippet
if face_match:
    mark_attendance(name)
    arduino.write(b'1') # Send signal to blink LED
else:
```

```
print("Unknown Face")
```

5. Key Calculations

The system uses Cosine Similarity to verify faces. If the result $D \leq \text{threshold}$, the face is considered a match. Typically, for DeepFace with VGG-Face model, a threshold of **0.40** is used for optimal accuracy.

6. Troubleshooting & Tips

- **Lighting:** Ensure the room is well-lit for the camera to detect features clearly.
- **Serial Port:** Always check which COM port your Arduino is connected to (e.g., COM3 or COM4) and update it in the Python code.
- **Excel Errors:** Ensure the Excel file is closed while the program is running, or Pandas won't be able to write data to it.

7. Conclusion

This system eliminates manual attendance marking, reducing errors and saving time. By combining the power of AI (Software) with the tangibility of Arduino (Hardware), we create a robust, professional-grade solution.